

Open Project
CIG, IIT Roorkee

AI-POWERED EVENT MEDIA MANAGEMENT PLATFORM

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Tech Stack:

React.js | Node.js | PostgreSQL | AWS

Rekognition | Cloudinary

PROBLEM STATEMENT

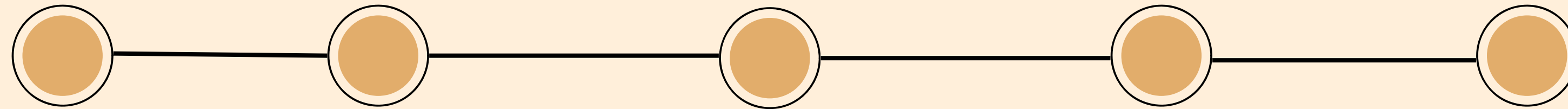
During college events, hundreds of photos are captured and shared across different platforms. It becomes difficult for participants to find their own pictures, and event organizers also face challenges in managing and distributing media efficiently.

The main goal of this project is to create a single platform where event photos can be uploaded, organized, searched, and accessed easily.

PROJECT OBJECTIVES

- Create a centralized platform for event media.
- Allow users to upload and manage event photos.
- Help users find their own photos using face recognition.
- Enable interaction through likes and comments.
- Provide useful event statistics through a dashboard.

SYSTEM ARCHITECTURE



Frontend
(React.js)

Backend
(Node.js
+ Express)

PostgreSQL
Database

Cloudinary
(Image
Storage)

AWS
Rekognition
(Face Matching)

Deployment - Render(backend) and Vercel(frontend)

<https://event-media-platform-tazd.vercel.app>

<https://event-media-platform.onrender.com>

TECHNOLOGY STACK

- Frontend: React.js
- Backend: Node.js, Express.js
- Database: PostgreSQL
- Authentication: JWT

- Cloud Storage: Cloudinary
- Face Recognition: AWS Rekognition
- Charts & Analytics: Recharts
- Deployment: Vercel and Render

MAIN FEATURES

- User Registration and Login
- Event Creation and Management
 - Photo Upload and Gallery
 - Smart Search using Tags
- Search by User Name and Upload Date
- Like and Comment System
 - Analytics Dashboard
 - Watermarked Downloads
 - AI-based Face Recognition

HOME PAGE

EventHub

Events

Create Event

My Photos

Dashboard

Login

Signup

Event Media Platform

Create, manage and share event memories

Search events...



gsv

avg

ww

private

01/06/2026

Open Gallery →



hjbn

nj

jn

public

13/05/2026

Open Gallery →



wrwt

124357

gsgs

public

10/02/9337

Open Gallery →

CREATE EVENT

EventHub

EventsCreate EventMy PhotosDashboardLoginSignup

Create Event

Title

Choose file No file chosen

Description

Category

dd/mm/yyyy

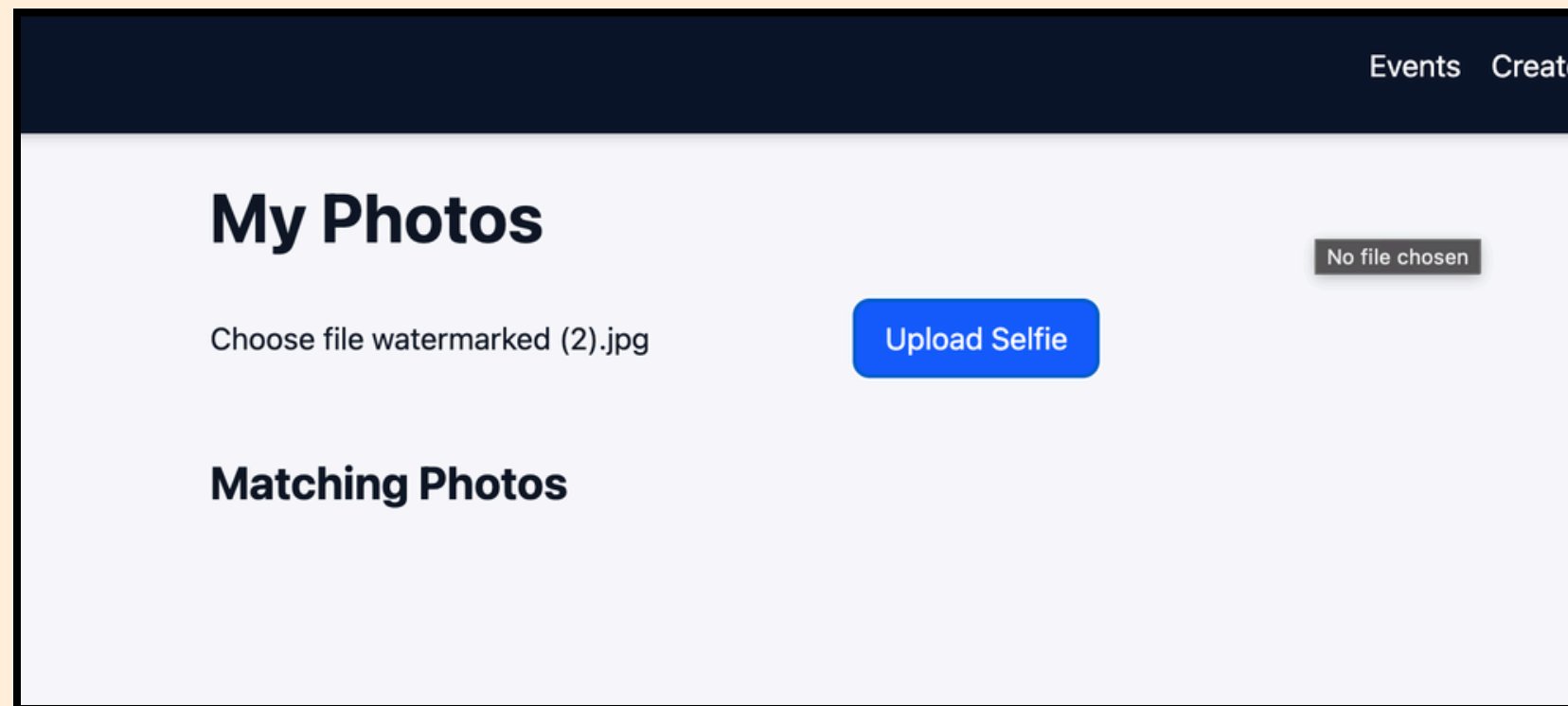
Public

Create Event

EventHub

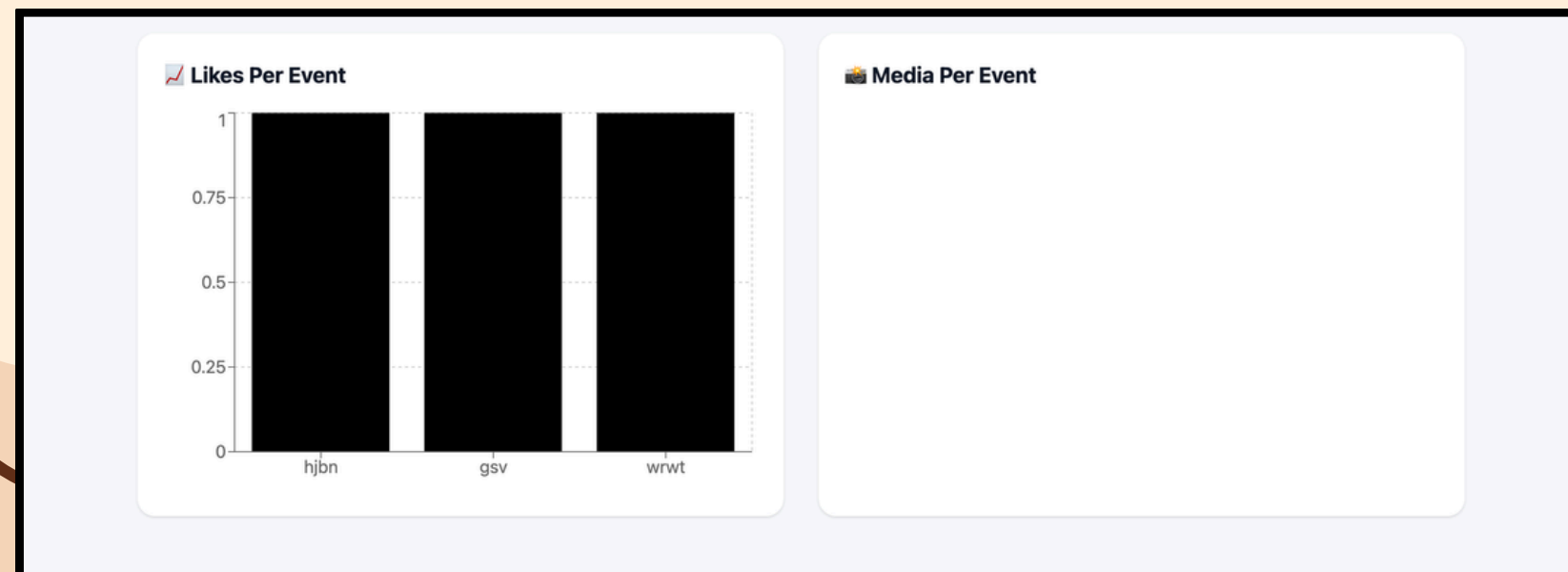
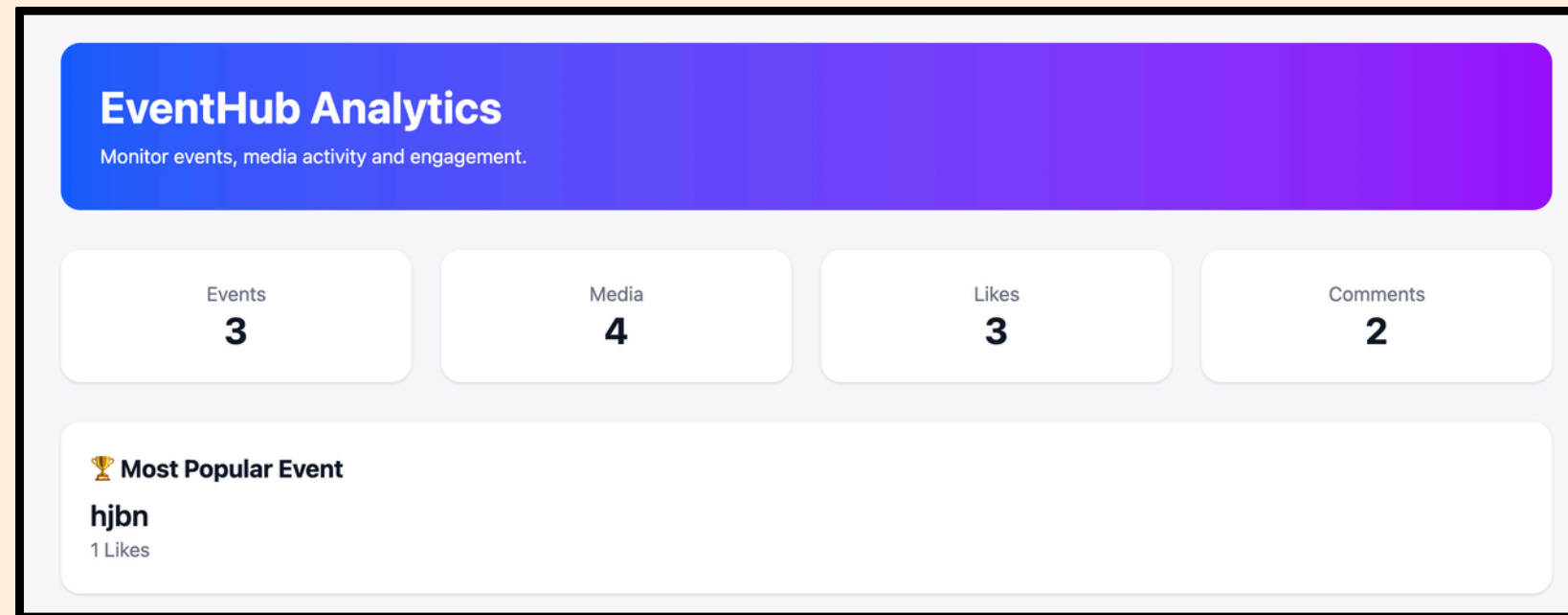
AI-Powered Event Media Platform

FACE RECOGNITION MODULE



- Users can upload a selfie to the platform.
- The system compares the uploaded selfie with all event images using AWS Rekognition and displays matching photos automatically.
- This removes the need to manually search through hundreds of images.

Analytics Dashboard



The dashboard provides a quick overview of platform activity.

It displays:

- Total Events
- Total Media Files
- Total Likes
- Total Comments
- Most Popular Event
- Likes per Event
- Media Distribution

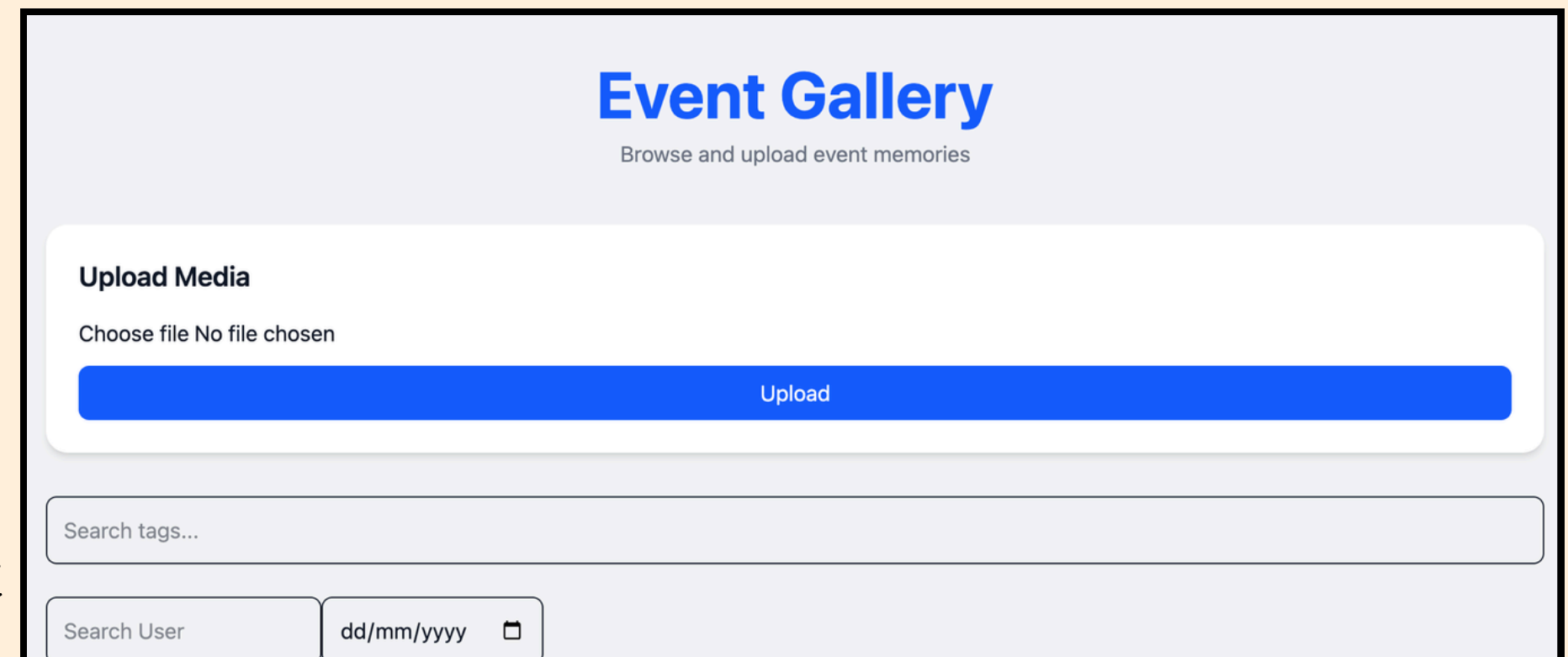
This helps organizers understand user engagement.

Search and Filtering

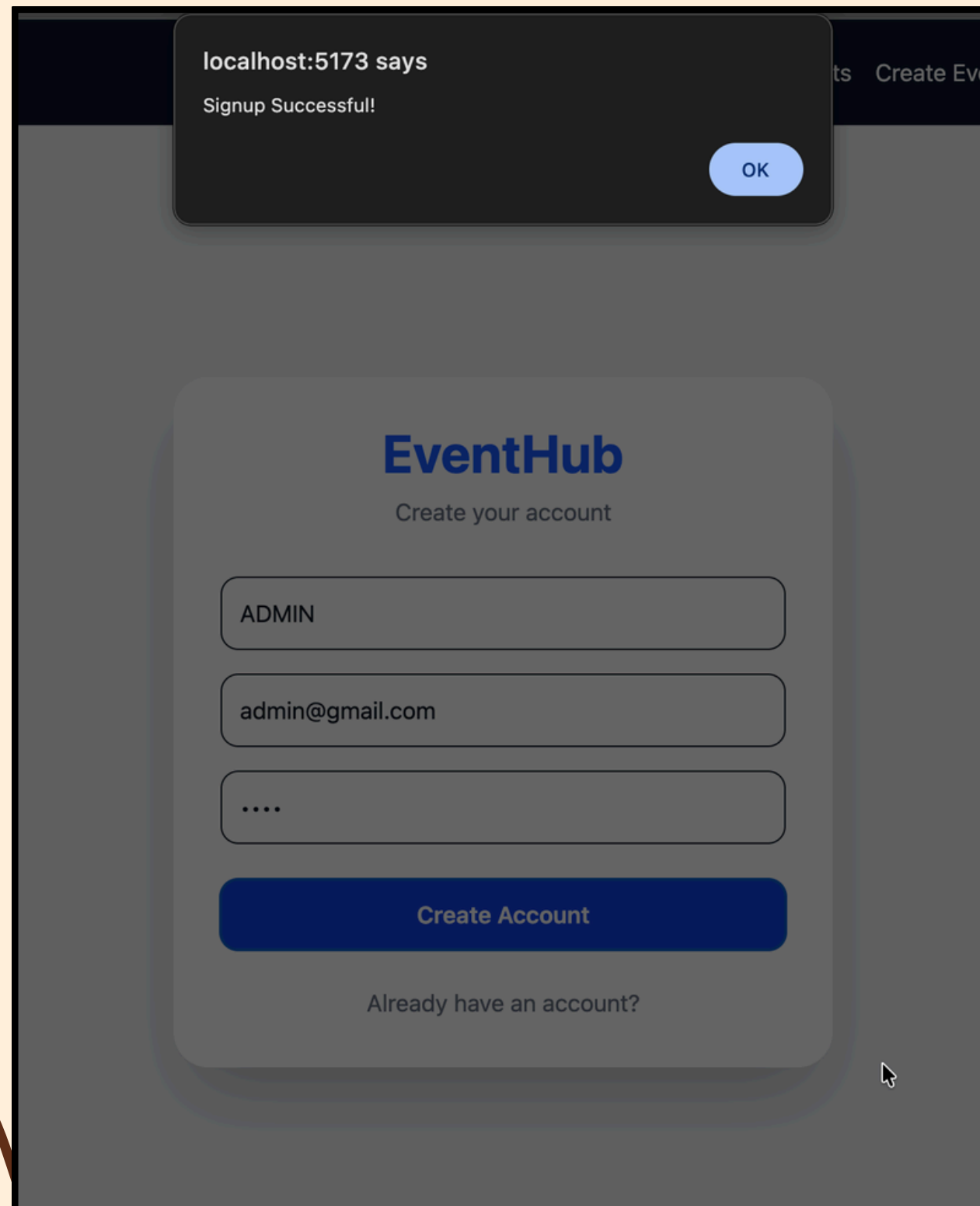
To improve user experience, multiple search options have been added:

- Event Name
- Tags
- User Name
- Upload Date

Users can quickly locate specific photos without scrolling through the entire gallery.

A mockup of an 'Event Gallery' web interface. The header is light blue with the title 'Event Gallery' in bold blue text and the subtitle 'Browse and upload event memories' in smaller grey text. Below the header is a white 'Upload Media' section containing a file selection area with the text 'Choose file No file chosen' and a blue 'Upload' button. At the bottom, there are three search filters: a large 'Search tags...' input field, a 'Search User' input field, and a date input field showing 'dd/mm/yyyy' with a calendar icon.

Security Features



The platform uses JWT authentication to protect user accounts.

Downloaded images also contain a dynamic watermark that includes event information and user role, helping prevent unauthorized sharing.

OUTCOME

- The project successfully combines web development with AI services to simplify event media management.
- It provides an easy way to upload, organize, search, and retrieve event photos while improving the overall user experience.

FUTURE IMPROVEMENTS

- Mobile application support
- Automatic AI-based image tagging
- Real-time notifications
- Better face recognition accuracy
- Social media sharing integration

THANK
YOU

